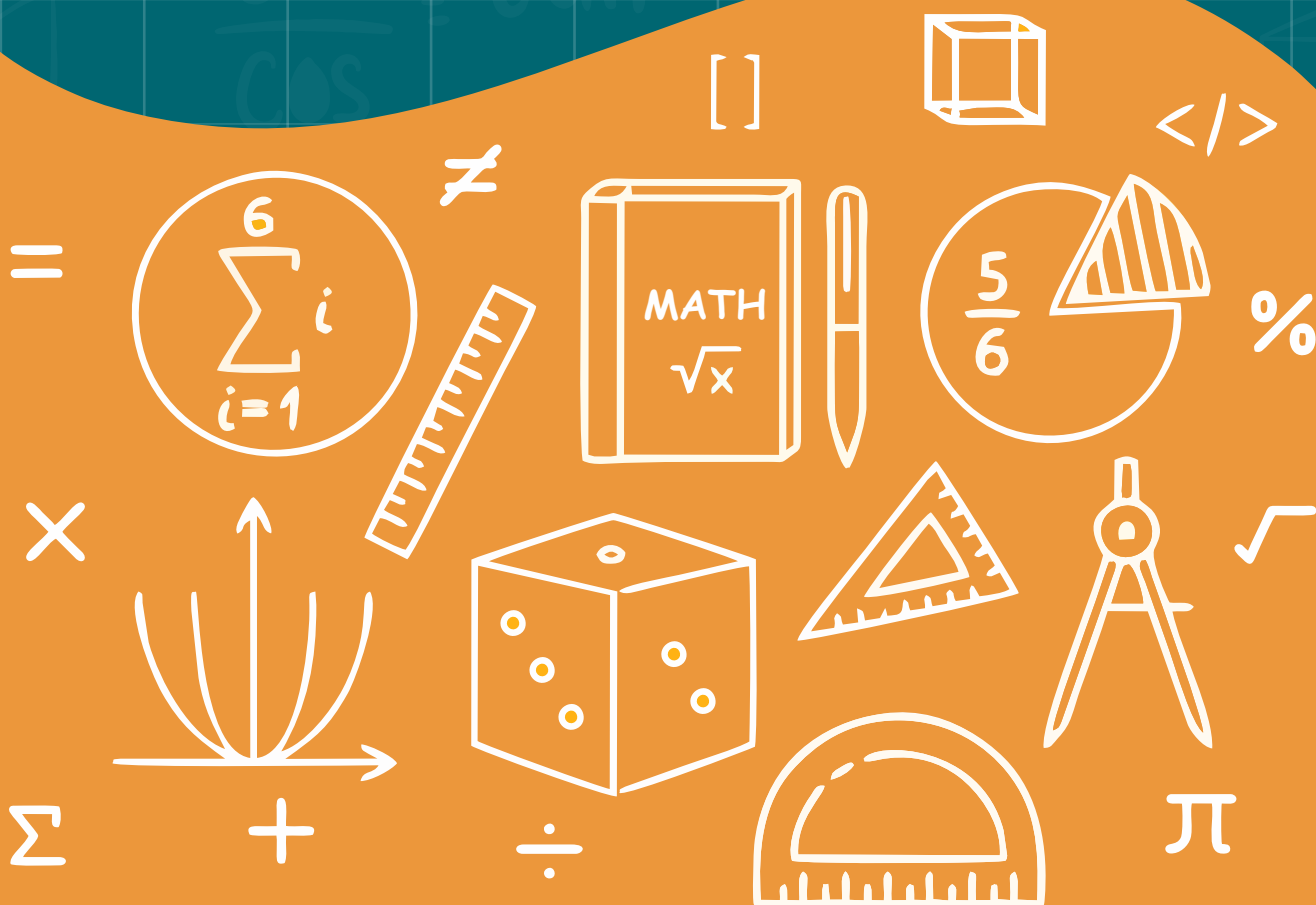


UNIT

9

LINEAR EQUATION



ALGEBRAIC EQUATION

An algebraic sentence involving the sign of equality “=” is called an algebraic equation or simply an equation e.g.

(i) $6 - 2 = 4$

(ii) $2x + 4 = 8$

are examples of an algebraic equation.

ALGEBRAIC EXPRESSION

An algebraic expression is a combination of numbers, variable and arithmetical operations.

Thus $2a + y$, $3l + z$, $2x - 3y$ are all examples of algebraic expressions.

Differentiate between equation and an expression:

Whereas two expressions connected by sign of equality is called an equation. In an equation, the two expressions are equated with the symbol “=”.

$2x + 3 = 9$, $x - y = 7$ are examples of equation.

Examples of expression:

(i) Four is added to twice of a number i.e. twice of a number plus four is: $2x + 4$.

(ii) Five is subtracted from thrice of a number i.e. thrice of a number minus five is: $3y - 5$.

(iii) Two is added to thrice of a number i.e. thrice of a number plus two is: $3n + 2$.

(iv) 6 is multiplied by a number i.e. six times of a number is: $6z$

(v) A number is divided by 8 i.e. eight part of a number is: $\frac{m}{8}$

Examples of equation:

(i) Four is added to twice of a number and the result is 10 i.e. $2x + 4 = 10$

(ii) Three is subtracted from thrice of a number and the result is 1.
i.e. $3y - 3 = 1$

(iii) Three multiplied by a number is 9 i.e. $3z = 9$

(iv) Two added to thrice of a number is 5 i.e. $3n + 2 = 5$

(v) A number divided by 8 is 3 i.e. $\frac{m}{8} = 3$

Example 1: Construct the linear equations in one variable of the following.

(i) Four times a number increased by 5 equals 11.

(ii) If a number is subtracted from another number which is 5 more than thrice the first number, the result is 31.

Solution:

(i) Let x be the number. Then four times of x is $4x$ and increasing this by 5 gives $4x + 5$ which is equal to 11. Hence the required equation is: $4x + 5 = 11$

(ii) Let x denote the number. Then thrice the number is $3x$. Now other number which is 5 more than $3x$ is $3x + 5$. The difference of $3x + 5$ and x is $(3x + 5) - x$ which is equal to 31. Hence the required equation is: $(3x + 5) - x = 31$

EXERCISE 9.1

Q1. Construct the following linear equations.

(1) The difference of a number and twenty is thirty two.

(2) The sum of a number and 8 is 15.

(3) Five multiplied by a number is 35.

(4) Sixteen divided by a number is equal to 4.

(5) A number increased by two is 4.

(6) Fifteen is decreased from a number, and the result is 5.

LINEAR EQUATION

The equation which contains one variable and its degree is also one. Such equation is called linear equation.

Examples of linear equations:

$$x = 4, x + 8 = 15, 5y = 30.$$

An open sentence which contains the symbol "=" is called equations. The value of the variable which makes the equation true, is called the solution of the equation.

Example 1: Solve $x + 5 = 11$

Solution:

$$x + 5 = 11$$

Subtracting 5 from both sides, we get.

$$x + 5 - 5 = 11 - 5$$

$$\text{or } x + 0 = 6$$

$$\text{or } x = 6$$

Same number can be subtracted from both sides of the equation.

Example 2: Solve $x - 8 = 21$

Solution: $x - 8 = 21$

Adding 8 to both sides we get

$$x - 8 + 8 = 21 + 8$$

$$\text{or } x = 29$$

$$\text{checking } x - 8 = 21$$

$$29 - 8 = 21 \quad (\text{by putting } x = 29)$$

$$\text{or } 21 = 21 \quad \text{which is a true sentence.}$$

Same number can be added to the both sides of the equation.

Example 3: Solve: $7x = 84$

Solution:

$$7x = 84$$

Dividing both sides by 7, we get

$$\frac{7x}{7} = \frac{84}{7}$$

$$x = 12$$

Both sides of equation can be divided by the same non-zero number.

Example 4: Solve $\frac{x}{3} = 7$

Solution:

$$\frac{x}{3} = 7$$

Multiplying both sides by 3, we get

$$\frac{x \times 3}{3} = 7 \times 3$$

$$x = 21$$

Both sides of equation can be multiplied by the same number.

Both sides of the equation can be interchanged

Example 5: Solve: $3t + 5 = 20$

Solution:

$$3t + 5 = 20$$

$$3t + 5 - 5 = 20 - 5 \quad (\text{subtracting 5 from both sides})$$

$$3t = 15$$

$$\frac{3t}{3} = \frac{15}{3} \quad (\text{Dividing both sides by 3})$$

$$t = 5$$

Example 6: Find the solution of the following equations:

(i) $\frac{1}{2}a + 5 = a - \frac{1}{3}$

(ii) $\frac{x+1}{2} = \frac{x+4}{3}$

(iii) $\frac{3x+4}{5+2x} = 1$

Solution:

(i) $\frac{1}{2}a + 5 = a - \frac{1}{3}$

Multiplying both sides by L.C.M 6.

$$6\left(\frac{1}{2}a + 5\right) = 6\left(a - \frac{1}{3}\right)$$

$$\cancel{6}^3 \left(\frac{1}{\cancel{2}} a \right) + 6(5) = 6(a) - \cancel{6}^2 \left(\frac{1}{\cancel{3}} \right)$$

$$3a + 30 = 6a - 2$$

$$30 + 2 = 6a - 3a$$

$$32 = 3a$$

$$\text{or } 3a = 32$$

$$a = \frac{32}{3}$$

$$\text{(ii)} \quad \frac{x+1}{2} = \frac{x+4}{3}$$

Multiplying both sides by L.C.M 6.

$$6 \left(\frac{x+1}{2} \right) = 6 \left(\frac{x+4}{3} \right)$$

$$3(x+1) = 2(x+4)$$

$$3x + 3 = 2x + 8$$

$$3x - 2x = 8 - 3$$

$$x = 5$$

$$\text{(iii)} \quad \frac{3x+4}{5+2x} = 1$$

Multiplying both sides by $(5 + 2x)$

$$(\cancel{5+2x}) \left(\frac{3x+4}{\cancel{5+2x}} \right) = (5+2x) 1$$

$$3x + 4 = 1(5 + 2x)$$

$$3x + 4 = 5 + 2x$$

$$3x - 2x = 5 - 4$$

$$x = 1$$

EXERCISE 9.2

Q1. Solve the following equations.

(i) $x - 7 = 8$

(ii) $y + 5 = 12$

(iii) $x + 2 = 8$

(iv) $7y = 14$

(v) $3x = 30$

(vi) $y - 16 = 2$

(vii) $\frac{x}{4} = 3$

(viii) $6x + 1 = 49$

(ix) $8x = 32$

(x) $7y + 2 = 44$

(xi) $6 = x + 9$

(xii) $9x - 21 = 78$

Q2. Solve for the variable.

(i) $3x - 3 = 24$

(ii) $3z - 2 = 16$

(iii) $3v - 6 = 3$

(iv) $3x + 10 = 31$

(v) $3u + 6 = 9$

(vi) $2z + 8 = 18$

(vii) $3v - 4 = 5$

(viii) $3z - 7 = 5$

(ix) $2x + 1 = 21$

(x) $2c - 1 = 3$

(xi) $2a - 10 = 2$

(xii) $2u + 6 = 16$

(xiii) $3z - 8 = 23$

(xiv) $3x - 9 = 21$

(xv) $2u - 3 = 5$

Q3. Evaluate the following expressions.

When $x = 3$

$x - 2 =$ _____

$11 + x =$ _____

$4 + x =$ _____

$x - 9 =$ _____

$x - 3 =$ _____

$x + 12 =$ _____

$20 - x =$ _____

$13 - x =$ _____

$x - 16 =$ _____

When $x = 7$

$14 + x =$ _____

$x - 11 =$ _____

$x + 17 =$ _____

$x + 1 =$ _____

$x - 4 =$ _____

$x - 9 =$ _____

$13 + x =$ _____

$11 - x =$ _____

$5 + x =$ _____

When $x = 13$

$x + 7 =$ _____

$8 - x =$ _____

$13 + x =$ _____

$6 + x =$ _____

$x - 14 =$ _____

$x + 5 =$ _____

$10 - x =$ _____

$x - 9 =$ _____

$15 - x =$ _____

$6 + X = \underline{\hspace{2cm}}$

$14 - X = \underline{\hspace{2cm}}$

$X + 7 = \underline{\hspace{2cm}}$

$16 + X = \underline{\hspace{2cm}}$

$X - 6 = \underline{\hspace{2cm}}$

$8 - X = \underline{\hspace{2cm}}$

$X + 2 = \underline{\hspace{2cm}}$

$X - 5 = \underline{\hspace{2cm}}$

$X - 12 = \underline{\hspace{2cm}}$

$X + 6 = \underline{\hspace{2cm}}$

$3 - X = \underline{\hspace{2cm}}$

$X - 15 = \underline{\hspace{2cm}}$

$8 + X = \underline{\hspace{2cm}}$

$X + 21 = \underline{\hspace{2cm}}$

$12 + X = \underline{\hspace{2cm}}$

$X - 3 = \underline{\hspace{2cm}}$

$1 - X = \underline{\hspace{2cm}}$

$X + 4 = \underline{\hspace{2cm}}$

WORD PROBLEMS

1. Sarah stitches 44 mirror pieces on a red and green shirt. If 6 more mirror pieces are stitched on red shirt than green shirts, find the mirrors stitched on green shirt.
2. In a cricket match Jamil and Saleem enhanced the score of the team by 84 runs. If Jamil scored 12 runs more than Saleem. Find the score of Saleem.
3. The sum of a number and its double is 9. Find the number.
4. The product of 7 and the sum of a number and 6 is 77. Find the number.
5. The sum of 6 and four times a number is 42. Find the number.